Federated learning for cross-bank fraud detection

"What studies apply federated learning or federated graph neural networks to financial fraud detection across multiple banks? What accuracy, privacy and communication problems do they report?"
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What the studies show

Several papers apply **federated learning (FL)** or **federated graph neural networks (federated GNNs)** to financial fraud detection across institutions, and they consistently report the same three themes: **good detection performance**, **better privacy than centralized training**, and **communication/coordination costs that become a practical bottleneck** [118] [119] [120] [121] [122].

Main studies and what they report

Study	Approach	Reported performance	Privacy / communication issues
2SFGL: A Simple And Robust Protocol For Graph-Based Fraud Detection (2022)	Federated graph learning with "virtual fusion" of multi-party graphs, then GCN or GraphSAGE training [118]	GCN + 2SFGL improved performance by **17.6%–30.2%** over FedAvg; GraphSAGE + 2SFGL improved by **6%–16.2%** over FedAvg [118]	Designed specifically because institutions cannot share raw data; the paper frames FL as the privacy-preserving alternative [118]
Credit card fraud detection based on federated graph learning (2024)	Federated graph learning framework for credit-card fraud [119]	On CreditCard and IEEE-CIS datasets, it improved **Recall, F1, and AUC** over other FL models, but **accuracy and precision were sometimes worse** than the strongest baselines [119]	Centralized baselines were described as impractical because of **data security concerns**; the federated setup preserves privacy but trades off some precision [119]
Fed-RD: Privacy-Preserving Federated Learning for Financial Crime Detection (2024)	FL with differential privacy and secure aggregation [3]	Reported **high accuracy** with **minimal degradation as privacy increased** [3]	Explicitly uses **differential privacy** and **SMPC** to protect model updates; privacy comes with extra protocol complexity [3]
A Privacy-Preserving Federated Learning Model for Money Laundering Detection (2025)	FL with DP and secure aggregation [3]	Reported **accuracy up to 85.71%** [3]	Again emphasizes privacy-preserving training, but does not remove the cost of secure coordination [3]

Study	Approach	Reported performance	Privacy / communication issues
TrustFed-HE (2026)	Federated fraud analytics with homomorphic secure aggregation and poisoning-resistant training [3]	Reported **high AUC-PR/recall** in the excerpt [3]	Focuses on robustness to **poisoning** and privacy-preserving aggregation, implying higher computational burden from secure aggregation [3]
FedFairGNN (2026)	Federated GNN for fraud detection with differential privacy and fairness constraints [123]	Demonstrated a **competitive performance-fairness trade-off** and reduced demographic disparity [123]	Highlights FL-specific graph problems: **graph heterogeneity across clients** and **bias propagation**; uses DP on gradients to protect sensitive information [123]
A federated graph neural network approach to financial fraud detection (2026)	Privacy-preserving federated heterogeneous GNN anti-fraud architecture [124]	Improved **AUPRC to 0.932**, with **31.2 ms** single-node inference latency [124]	Claims to block topology inference attacks and reduce bandwidth overhead **by nearly an order of magnitude** per round, showing communication is a major design target [124]
HiFraud (2026)	Hierarchical privacy-preserving FL for cross-institutional fraud detection [120]	**AUC-ROC 0.935** under $\varepsilon = 2.3$, **convergence reduced from 49 to 30 rounds** [120]	Reports that uniform DP hurts sparse fraud signals; uses hierarchical clustering and privacy-adaptive aggregation to balance privacy and utility [120]
FedRisk (2026)	Explainable federated framework for cross-institutional risk control [125]	**AUC-ROC 0.912**, only **1.8% below centralized GBDT**; **99% reduction** in cross-institutional data transmission [125]	Addresses **non-IID data**, privacy, and communication with adaptive DP and secret sharing; achieves low bandwidth per round [125]
FedAugSec (2026)	Federated learning with selective encryption and data augmentation [121]	Experimental results show feasibility and effectiveness in fraud detection [121]	Explicitly targets the two main bottlenecks: **secure parameter transmission** and **communication overhead** under non-IID shifts [121]

Accuracy patterns

The evidence suggests FL and federated GNNs can reach **strong fraud-detection performance**, but the best metric varies by paper and setup [118] [119] [120] [124] [125].

- Some graph-federated systems improve **recall and AUC** more than raw accuracy, which is important in rare-fraud settings [119].
- Other systems report near-centralized performance while preserving privacy, such as **AUC-ROC 0.935** for HiFraud and **AUC-ROC 0.912** for FedRisk [120] [125].
- Several papers report large gains versus FedAvg or centralized baselines, including **17.6%–30.2%** improvements in the 2SFGL study and **AUPRC 0.932** in the federated GNN architecture [118] [124].

Privacy problems they report

The privacy story is generally positive, but not free:

- **Raw data stays local**, which is the main reason FL is used at all [118] [119].
- Many systems add **differential privacy**, **secure aggregation**, **homomorphic encryption**, or **secret sharing** to protect updates from leakage [3] [120] [121] [123] [125].
- The papers also warn that FL is still exposed to **poisoning**, **gradient leakage**, and **membership inference** unless extra defenses are added [3] [120] [125].
- FedRisk reports membership-inference resistance but still treats privacy as a tunable budget, not a solved problem [125].

Communication problems they report

Communication is the most consistent operational bottleneck across the evidence:

- FL requires repeated exchange of **model updates**, which creates **round-based bandwidth cost** [121] [125].
- The 2026 federated GNN architecture explicitly says it reduced **bandwidth overhead per round by nearly an order of magnitude**, showing communication is a key constraint in graph-based collaboration [124].
- HiFraud shortened convergence from **49 to 30 rounds**, which matters because fewer rounds means less communication [120].
- FedRisk reduced transmission by **99%**, again indicating that communication efficiency is central to making cross-bank collaboration practical [125].
- The broader FL review says **high communication cost** and **data/device heterogeneity** are fundamental obstacles in FL generally [122].

Bottom line

The strongest cross-bank fraud-detection results come from **federated graph learning** and **privacy-preserving FL variants**, especially when they combine graph structure with secure aggregation or differential privacy [118] [119] [120] [124] [125]. The main trade-off is that better privacy usually means more protocol complexity and more communication overhead, while graph heterogeneity and non-IID bank data make training and aggregation harder [120] [121] [122] [123].

Confidence level: High — multiple independent studies across 2022–2026 consistently report the same performance/privacy/communication trade-offs, though reported metrics differ by dataset and model design.

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